

CREDIT CARD

APPROVAL PREDICTION

Abstract :

The Credit Card Approval Prediction System is a web-based application developed to automate the credit card approval process using Machine Learning and rule-based underwriting. The system analyzes applicant information such as age, annual income, employment experience, CIBIL score, debt, and previous credit history to determine whether a credit card application should be approved or rejected. By combining a trained Machine Learning model with predefined banking rules, the application provides accurate, consistent, and transparent decision-making while reducing manual effort and processing time.

The application is built using Python, Flask, HTML, CSS, and JavaScript, with Scikit-learn, XGBoost, Pandas, and NumPy used for data preprocessing, model training, and prediction. It includes separate interfaces for customers, analysts, and compliance officers, enabling users to submit applications, review prediction results,

and verify compliance with banking policies. The trained model is stored using Joblib for efficient and reusable predictions.

Overall, the Credit Card Approval Prediction System demonstrates the practical application of Machine Learning in the banking and financial sector by improving the efficiency, accuracy, and reliability of credit card approval decisions. It serves as a scalable and user-friendly solution suitable for academic projects and real-world financial applications.

Problem Statement:

The manual credit card approval process is time-consuming, inconsistent, and prone to human error, as it requires the evaluation of multiple applicant factors such as income, credit score, employment history, debt, and previous credit records. These challenges can result in delayed decisions and inaccurate approvals or rejections. This project addresses the problem by developing a Credit Card Approval Prediction System that uses Machine Learning and rule-based underwriting to automate the evaluation process, improve prediction accuracy, reduce processing time, and support reliable,

transparent, and efficient credit card approval decisions.

Features :

The Credit Card Approval Prediction System provides an automated and user-friendly platform for evaluating credit card applications using Machine Learning and rule-based underwriting. The key features of the system are as follows:

1. Customer Application Portal

- Provides a simple web interface for customers to submit credit card applications.
- Allows users to enter personal and financial details required for evaluation.

2. Machine Learning-Based Prediction

- Uses a trained Machine Learning model to predict whether a credit card application should be approved or rejected.
- Analyzes applicant data to generate accurate and consistent predictions.

3. Rule-Based Underwriting Engine

- Applies predefined banking rules to validate applications before generating the final decision.

- Ensures that approval decisions comply with standard banking policies and eligibility criteria.

4. Applicant Risk Assessment

- Evaluates the applicant's financial profile based on factors such as income, credit score, debt, and employment information.
- Classifies applications according to their level of risk.

5. Analyst Dashboard

- Enables analysts to review submitted applications and prediction results.
- Provides a centralized interface for monitoring customer applications.

6. Compliance Dashboard

- Allows compliance officers to verify approval decisions.
- Displays the underwriting rules applied during the evaluation process.
- Helps ensure transparency and regulatory compliance.

7. Automated Decision Making

- Reduces manual intervention by automatically processing applications and generating instant approval or rejection decisions.

8. Model Persistence

- Stores the trained Machine Learning model using Joblib, allowing the application to make predictions without retraining the model every time.

9. User-Friendly Web Interface

- Developed using HTML, CSS, and Flask to provide an interactive and easy-to-use interface for different users.

10. Fast and Reliable Processing

- Processes applicant information quickly, reducing the overall time required for credit card approval decisions while improving consistency and accuracy.

Tech Stack :

The Credit Card Approval Prediction System is developed using a combination of web development technologies and Machine Learning tools to ensure

efficient, accurate, and user-friendly credit card approval prediction.

Python is used as the primary programming language for implementing the application's backend logic, data processing, and Machine Learning model. It provides a simple and powerful environment for developing intelligent applications.

The Flask framework is used to build the web application. It handles user requests, connects the frontend with the backend, processes application data, and displays the prediction results through a web interface.

For Machine Learning, Scikit-learn is used to preprocess the dataset, train the prediction model, and evaluate its performance. The XGBoost algorithm is implemented to improve the accuracy and reliability of credit card approval predictions.

Pandas is used for loading, cleaning, and manipulating the dataset, while NumPy supports numerical computations and efficient handling of data arrays.

The trained Machine Learning model is saved and loaded using Joblib, allowing the application to make predictions without retraining the model every time the application starts.

The user interface is developed using HTML and CSS. HTML is used to design the structure of the web pages, while CSS enhances the appearance and provides a responsive and user-friendly layout. Jinja2, Flask's templating engine, is used to dynamically render web pages with prediction results.

The project uses a CSV dataset containing applicant information for training and testing the Machine Learning model.

The application can be developed and tested using Google Colab for model training and Visual Studio Code (VS Code) for web application development. The source code can be managed using Git and GitHub, enabling version control and easy collaboration.

Overall, this technology stack provides a robust, scalable, and efficient platform for developing a web-based Credit Card Approval Prediction System that delivers fast and accurate approval decisions.

Installation Steps :

1. Clone the Repository & Navigate to the Directory

```
cd credit card approval
```

2. Create and Activate a Virtual Environment

- On Windows:

```
python-m venv venv  
.\venv\Scripts\activate
```

On macOS/Linux:

```
python3-m venv venv  
source venv/bin/activate
```

3. Install Dependencies

```
pip install-r requirements.txt
```

4. Train/Verify the ML Model (Optional) If you need to regenerate the best_model.joblib binary file from the raw dataset:

```
python ml_pipeline.py
```

5. Run the Automated Tests Ensure the underwriting engine logic is functioning perfectly before starting the server:

```
pytest test_underwriting.py
```

6. Start the Web Server

```
python app.py
```

The application will be accessible at http://127.0.0.1:5000/ in your web browser.

Usage Instructions :

Usage Instructions

Follow the steps below to use the Credit Card Approval Prediction System.

Step 1: Start the Application

Run the application using the following command:

```
python app.py
```

Open your web browser and go to:

<http://127.0.0.1:5000>

Step 2: Open the Customer Page

Open the Customer Portal from the home page.

Step 3: Enter Applicant Details

Fill in all the required information, such as:

- Age
 - Annual Income
 - Employment Experience
 - CIBIL Score
 - Debt
 - Previous Credit Default
 - Other required details
-

Step 4: Submit the Application

Click the Predict button after entering all the details.

Step 5: View the Result

The system will process the data and display whether the credit card application is Approved or Rejected.

Step 6: Review Applications

The Analyst Dashboard allows analysts to review submitted applications and prediction results.

Step 7: Check Compliance

The Compliance Dashboard displays the approval decision and verifies whether the application follows the underwriting rules.

Step 8: Exit the Application

After completing the process, close the browser and stop the application by pressing Ctrl + C in the terminal.

Expected Output

The system analyzes the applicant's information using Machine Learning and predefined banking rules, then

displays the final credit card approval or rejection result quickly and accurately.

Future Scope :

The Credit Card Approval Prediction System can be enhanced in the future by adding more advanced features and integrating modern banking technologies. Some possible future improvements are:

- Integrate the system with real-time banking and credit bureau databases for more accurate credit card approval decisions.
- Implement user authentication and role-based access to provide secure login for customers, analysts, and administrators.
- Develop a mobile application for Android and iOS to allow users to apply for credit cards from their smartphones.
- Add email and SMS notification services to inform applicants about their application status instantly.
- Integrate Aadhaar, PAN, and other identity verification services to automate customer verification.

- Improve prediction accuracy by training the model with larger and real-world banking datasets.
- Incorporate Explainable Artificial Intelligence (XAI) techniques to provide clear explanations for approval or rejection decisions.
- Deploy the application on cloud platforms such as AWS, Microsoft Azure, or Google Cloud for better scalability and availability.
- Develop an admin dashboard with detailed reports, analytics, and graphical visualizations to monitor application statistics and system performance.
- Extend the system to support other financial services such as loan approval, insurance eligibility prediction, and fraud detection.

Overall, these enhancements will improve the system's accuracy, security, scalability, and usability, making it more suitable for real-world banking and financial applications.